Pergamon

0273-1177(95)00156-5

Adv. Space Res. Vol. 16, No. 7, pp. (7)181–(7)184, 1995
Copyright © 1995 COSPAR
Printed in Great Britain. All rights reserved.
0273–1177/95 $9.50 + 0.00

SPACE FLIGHT DATA FROM THE ISOTHERMAL DENDRITIC GROWTH EXPERIMENT

M. E. Glicksman,* M. B. Koss,* L. T. Bushnell,*
J. C. LaCombe* and E. A. Winsa**

** Materials Engineering Department, Rensselaer Polytechnic Institute, Troy, NY 12180-3590, U.S.A.*
*** MS500-115, Space Experiments Division, NASA Lewis Research Center, Cleveland, OH 44135, U.S.A.*

ABSTRACT

The Isothermal Dendritic Growth Experiment (IDGE) is a NASA space flight experiment which flew as part of the United States Microgravity Payload (USMP-2), in early 1994. The IDGE measured dendritic growth rates, tip radii, and crystal morphologies of ultra-pure succinonitrile [CN-$(CH_2)_2$-CN] at supercoolings in the range from 0.05 - 2.0 K. Data taken in the form of slow-scan binary digitized images telemetered to the ground from orbit in near-real time, combined with the IDGE terrestrial data set, provide the first quantitative assessment of various theories on dendritic solidification and the effects of convection on dendritic growth.

INTRODUCTION AND BACKGROUND

Dendritic growth is the ubiquitous form of crystal growth encountered when metals, alloys, and many other materials solidify under low thermal gradients, a situation which typically occurs in most industrial solidification processes. A clear elucidation of dendritic growth kinetics under well-characterized diffusion-controlled conditions is crucial in order to achieve a rigorous test of dendritic growth theories, and, ultimately, to predict and achieve desired microstructures and hence, physical properties, in a variety of solidification processes.

The growth of dendrites in pure liquids is generally acknowledged to be controlled by the diffusive transport of latent heat from the moving crystal-melt interface as it advances into its supercooled melt. Under terrestrial conditions this flow of heat, particulary at small supercoolings, and the resulting dendritic growth speed and size is greatly affected by natural convection. A number of theories of dendritic crystal growth, based on various transport mechanisms, physical assumptions, and mathematical approximations, have been developed over the last forty years (see the recent review by one the authors [1]). A mathematical solution to the dendritic heat conduction problem was first analyzed by Ivantsov [2], who modeled the dendrite as a paraboloidal body of revolution, growing at a constant velocity, V. The resultant thermal diffusion field can be expressed mathematically and exactly in paraboloidal coordinates moving with the dendritic tip.

This diffusion solution is, however, incomplete, insofar as it specifies the tip growth Peclet number, $VR/2\alpha$, (where R is the tip radius of curvature, V the tip velocity, and α is the thermal diffusivity) as a function of the supercooling ΔT. The explicit dynamic operating state, V, and R, is not explicitly specified. The Peclet number obtained from the Ivantsov solution for each specified supercooling, yields instead an infinite manifold of V, and R, ordered pairs (V,R), that satisfy the diffusion solution at that particular value of ΔT. One finds experimentally that unique, steady, operating states (V_{exp}, R_{exp}) are observed at any specified supercooling, ΔT. Considerable theoretical efforts within the physics community have been directed to answering the question as to whether and under what conditions a second equation or length scale exists, which, when combined with the Ivantsov diffusion solution, selects the unique (i.e., observed) dynamic operating state (see Ref. [1,3-6]). Although the underlying physics for these "theories of the second length scale" might in fact be quite different, their results are often encapsulated with a scaling constant that is defined as proportional to $1/VR^2$.

A great deal of prior experimental work has been performed using succinonitrile (SCN: the chemical formula for which is NC-$(CH_2)_2$-CN) as a model dendritic growth system because of its conveniently low melting

temperature, optical transparency, and well-characterized thermophysical properties [6,7]. To date, the large body of experimental evidence gathered on SCN and other suitable test materials, neither serves to confirm or deny any of the "selection rule" theories in combination with the Ivantsov diffusion model. The experimental situation, simply stated, is that there appears to be too narrow a range of supercoolings in any crystal-melt system thus far studied terrestrially that remains *both* free of convection effects, and permits the accurate determination of the dendrite tip radius of curvature. There has been some analysis estimating the effects of natural or forced convection on dendritic growth [8-10], but such analysis is usually coupled to a particular dendritic growth theory, and, consequently can not provide an independent test of theory. Thus, a low-gravity dendritic growth experiment was proposed to test definitively the kinetics of the diffusive dendrite growth problem, and to measure the effects of natural convection. Such a test requires that a single, clearly defined experimental parameter, viz., the supercooling, fully determine the resulting steady-state dendritic growth velocity, V, and tip size, R. The presence of a body force, like gravity, must be eliminated, because the vector field would cause the orientation of the growth axis to affect the outcome. i.e., V and R would depend on the spatial orientation of the growth.

EXPERIMENTAL RESULTS AND DISCUSSION

The Isothermal Dendritic Growth Experiment (IDGE) was launched on March 4, 1994, to a low-Earth circular orbit of approximately 163 nautical miles in the payload bay of the space shuttle Columbia (STS-62). The IDGE, one of four experiments comprising the second United States Microgravity Payload Mission (USMP-2) was designed, built, and operated, to satisfy these exacting scientific requirements by measuring the growth kinetics and morphology (tip speed and size) of SCN dendrites growing under microgravity conditions, with its concomitantly greatly reduced natural convection. A detailed description of the growth chamber and the IDGE space flight hardware has been published previously [11,12].

Although the complete analysis of the on-orbit scientific results will have to await the return of the more than 400 optical photomicrographs taken as 35mm negatives, we have already analyzed most of the binary images taken during each dendritic growth cycle. These binary images were telemetered to the Payload Operations and Control Center (POCC) at the Marshall Space Flight Center (MSFC) and then interpreted by the IDGE science teams assembled at the POCC, in Huntsville, AL, and at Rensselaer Polytechnic Institute, in Troy, NY. Fig. 1 shows, as an example, four binary images from both the microgravity and terrestrial experiments. Fig 1a and 1b are for dendritic growth at 0.1 K supercooling in terrestrial gravity and microgravity respectively, and fig. 1c and 1d are for dendritic growth at 1.0 K supercooling also in terrestrial gravity and microgravity respectively. For each dendritic growth cycle, a binary image was captured and telemetered to Earth every 90 seconds. At the lower supercoolings, where the growth velocity was slower, we obtained more binary images. Dendrite tip velocities were calculated from the slope of a displacement-time plot made from the series of binary images. Fig. 2 shows one such time-displacement plot at of ca. 0.18 K supercooling. The analysis of the telemetered binary images was limited to dendrite growth velocity measurements only, as extraction of accurate dendritic tip radii

Figure 1: Slow Scan Television binary digitized images. Pair a) and b) 0.1 K supercooling at terrestrial gravity and microgravity; Pair c) and d) 1.0 supercooling at terrestrial and microgravity.

measurements demand a spatial resolution that is available in the 35mm film, but not in the binary images. However, the velocity measurements performed in near-real-time yielded results with sufficient accuracy and precision to permit definitive conclusions to be reached about convective effects on dendritic growth.

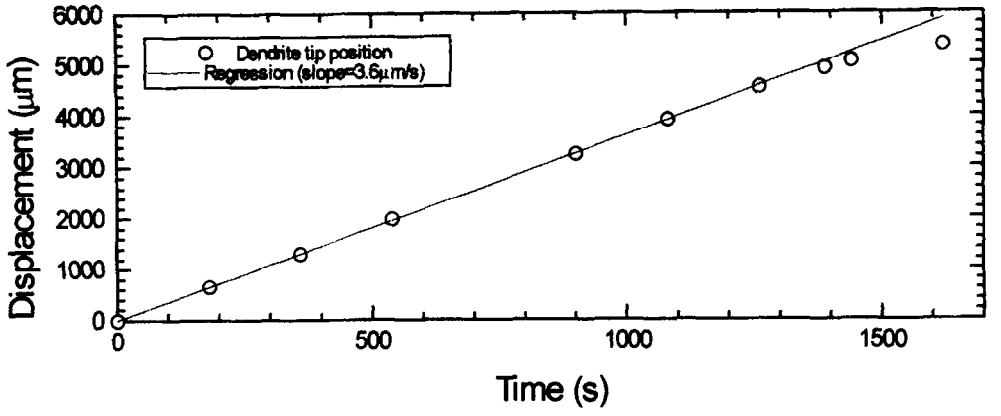

Figure 2: Time-displacment plot at ca. 0.18 K supercooling measured from series of binary images telemetered from the space shuttle Columbia (STS-62).

Fig. 3 contains a plot of three sets of dendritic growth data: 1) IDGE velocities measured terrestrially with the flight instrument, using the same SCN sample that was flown; 2) the terrestrial data set of Huang/Glicksman [7]; and 3) the on-orbit velocity data from USMP-2 that we obtained from the binary images. In addition, we show as the dotted curve, a theoretical velocity calculation generated from the Ivantsov diffusion solution, combined with a scaling constant, σ^*, adjusted to fit the microgravity velocity measurements over the upper range of these experiments where diffision-limited growth seems most likely.

The two ground-based data sets are in close agreement over the entire supercooling range. Thus the differences between these two data sets, arising from different chamber size and shape, different SCN specimen, and different eulerian angles of the [100] growth direction with respect to gravity (within 45 degrees of the 1 g_o vector), have little, if any, effect on these velocity data.

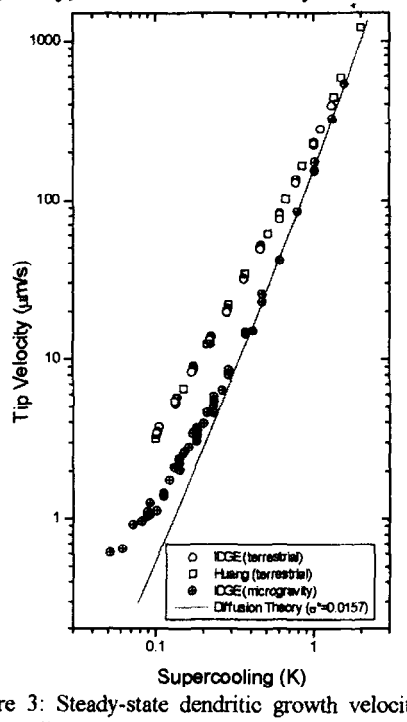

Figure 3: Steady-state dendritic growth velocity versus supercooling in both terrestrial gravity and microgravity.

However, Fig. 3 shows dramatic differences, particularly at the lower supercoolings, between terrestrial and microgravity conditions. At about 1.6K supercooling there appears to be a small reduction in the growth velocity under microgravity conditions as compared to terrestrial conditions. This reduction in velocity is clear at 1.3 K where the velocity measured in microgravity is 27% lower than the terrestrial data. Clearly, these particular results show that there are substantial convective effects, at 1 g_o, even at the higher supercoolings.

One can also see, in Fig. 3, that using one adjustable parameter, σ^*, with the Ivantsov diffusion solution, can not fit the microgravity velocity measurements over the entire range of supercooling. However, this method does yields velocity calculations in good agreement with the microgravity data over range of supercoolings of 0.4K and above. Thus, even in microgravity, at suffiently low supercoolings, convection effects substantially raise the growth velocity from what is expected by heat conduction alone.

The agreement between the dendritic velocity data in microgravity, and those velocities calculated with a one-parameter fit, is a necessary, but not sufficient condition, to claim that diffusion theory, combined with a scaling rule, quantitatively describe dendritic growth in microgravity. For complete evidence in support of this type of theory, the measured dendritic tip radii, R_{exp}, must also be consistent with the tip radius prediction. We can already infer from the observed agreement between the microgravity growth velocity data, and the one-parameter theoretical fit to those data, that the diffusion solution provides the proper functional dependence on supercooling. We do not, however, know from that fact if $\sigma^*=2\alpha d_0/VR^2$ is constant over the supercooling range. Thus, the observed differences between terrestrial and microgravity dendritic velocity data are due to convection. In addition, we speculate that the growing deviation of the velocity data from diffusion theory, at supercoolings less then 0.4K, arises from the increasing importance of convective effects. Microgravity experiments, after all, are not conducted at zero gravity, which is required by pure diffusion theory.

SUMMARY AND CONCLUSIONS

We measured the dendritic growth velocity of succinonitrile in microgravity using the IDGE instrument mounted on the USMP-2 platform in the payload bay of the space shuttle Columbia (STS-62). Data telemetered to the ground were analyzed, and when compared to terrestrial dendritic growth data, demonstrate that: 1) convective effects under terrestrial conditions are dramatic at the lower supercoolings, and remain significant even up to 1.3K supercooling, and, perhaps, above; 2) a diffusion solution to the dendrite problem, combined with a scaling constant, σ^*, yields growth velocity predictions consistent with the observed dendritic growth velocities as a function of supercooling. We are currently analyzing the 35mm photographs obtained during this flight experiment to study the influence of convection on dendritic growth, and assess the applicibility of selection rule theories of dendritic growth under microgravity conditions.

ACKNOWLEDGEMENTS

Special thanks are due for the continuing interest, and financial support provided by the NASA, Life and Microgravity Sciences and Application Division, (Code U), Washington, DC, under Contract NAS3-25368, with liaison provided through the Lewis Research Center, Cleveland, OH.

REFERENCES

1. M.E. Glicksman and S.P. Marsh, "The Dendrite", in Handbook of Crystal Growth, , ed. D.J.T. Hurle, (Elsevier Science Publishers B.V., Amsterdam, 1993), Vol **b**, part b, p.1077.

2. G.P. Ivantsov, Dokl. Akad. Nauk, USSR, **58**, 56, (1947).

3. J.S. Langer and H. Muller-Krumbhaar, Acta Metall., **26**, 1681, 1689, and 1697, (1978).

4. D. Kessler, J. Koplik, and H. Levine, Phys. Rev. A, **34**, 4980 (1986).

5. Y. Miyata, M.E. Glicksman, and T.H. Tirmizi, J. Crystal Growth, **112**, 683, (1991)

6. M.E. Glicksman, R.J. Schaefer, and J.D. Ayers, Met. Trans. A, **7A**, 1747, (1976).

7. S.C. Huang and M.E. Glicksman, Acta Metall., **29**, 701, (1981).

8. R. Ananth, and W.N. Gill, Chem. Eng. Comm., **68**, 1, (1988)

9. R. Ananth, and W.N. Gill, J. Crystal Growth, **91**, 587, (1988)

10. R. Ananth, and W.N. Gill, J. Crystal Growth, **108**, 173, (1991)

11. M.E. Glicksman et al., Met. Trans. A, **19A**, 1945, (1988).

12. M.E. Glicksman, et al., Advances in Space Research, **11**, 53, (1991).